How do we get from Rohan's ask to a list each of us can defend?

Build 2's workbook: one page per stage, then the numbers passport, the run sheet, the challenges log and the AI log. Fill it in pen. It comes with you to Thursday's gate and Saturday's defence.

Every stage page runs the same way: the question, why a real team does the step, a gym worked beside your boxes, and the line that says you are ready for the next stage. **Why a gym?** A different business shows the method without giving away your answer. The gym has 1,000 members, and about 15 in 100 let their membership lapse each month. The front desk can call a member and offer a free coaching session. A wasted call, to someone who would have stayed, costs 1 unit; a missed lapse costs 4 units. Every gym number here is invented.

Which page answers which question, and who leads each stage?

Roles rotate by stage, so nobody owns all the code. Write a name in each box.

Page	Stage, and the question it answers	Day	Builder	Evaluator	Decision owner
1	0 The business: who pays, and what does a wrong call cost?	Tue			
2	1 The frame: what is decided, from what is known when?	Tue			
3	2 Requirements: what must the system do, and never do?	Tue			
4	3 Options: two ways to decide, and which do we choose?	Wed			
5	4 The POC: does our list beat doing nothing on the error bill?	Wed			
6	5 The architecture: what are the boxes, and one box inside?	Wed, Thu			
7	6 The MVP slice: what could Rohan use tomorrow?	Fri			
8	7 Present and defend: which sentence would Rohan act on?	Sat	Slide 1 is drawn on the day		

The builder writes the code or the page, the evaluator checks every number, and the decision owner says what it means for Rohan. Stages 0 to 2 are on paper with no AI; from stage 3, AI may write code and every use goes in the AI log.

Stage 0, the business: who pays, and what does a wrong call cost?

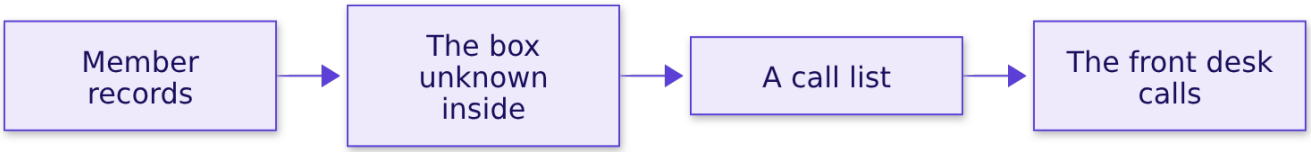
Who needs the answer. Rohan, before he trusts any list: a list that ignores who pays for its mistakes is one he cannot use.

The questions on the way. Who pays Kalpa, and for what? Who decides whom to call? What does each kind of wrong call cost? What goes into the box, and what comes out?

Why does a real team start here?

A list only matters if it changes what someone does, and it only costs money when someone pays for its mistakes. Ketan Doshi, who builds machine learning for enterprises, writes that in a demo someone has already framed the problem, while in a real project the team must find it and agree the business goal first (his Enterprise ML blog, 2021).

What does it look like for the gym, and for Kalpa?



Question	The gym (invented)	Kalpa: our answer, written in pen
Who pays, and for what?	Members pay a monthly fee to keep training.	
Who decides whom to call, and who calls?	The owner decides; the front desk calls.	
What does a wasted call cost, and why?	1 unit: a free session given to a member who would have stayed.	
What does a missed case cost, and why?	4 units: a member lost.	
What does doing nothing cost?	Every lapse is missed: 15 in 100 x 4 units = 60 units per 100 members.	
What goes into the box, and what comes out?	Member records in; a call list for the front desk out.	

Are we ready for stage 1?

You are ready for the next stage when every member can say, without notes and in under thirty seconds, who pays, who decides and what each wrong call costs.

Stage 1, the frame: what is decided, from what is known when?

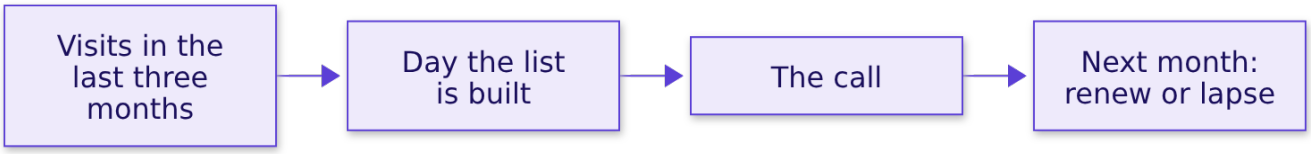
Who needs the answer. Rohan, who has to hear his problem said back correctly before he believes the group understood it.

The questions on the way. What exactly is decided, for whom and how often? What is known on the day of the decision, and what is not known yet? What outcome do we predict, and how often?

Why does a real team frame before it models?

A wireless company tested a retention offer and found that among its highest-risk customers, 95.3 percent left without the offer and 93.7 percent left with it, so calling the riskiest people barely changed anything (Ascarza, Journal of Marketing Research, 2018). "Who will leave?" and "whom will a call keep?" are different questions, and a team that does not choose answers the wrong one.

What does it look like for the gym, and for Kalpa?



Frame question	The gym (invented)	Kalpa: our answer, written in pen
What is decided?	Whether the front desk calls this member this month.	
About whom, and how often?	One member, once a month.	
What is known on the day of the list? Name the months.	Visits in each of the last three months, months as a member, plan type.	
What is not known yet, and when will it be?	Whether the member renews; known next month.	
What outcome do we predict?	Lapsed next month: yes or no.	
How often does it happen?	15 in 100 members: the base rate.	

Our frame in one sentence
We decide ___ for each ___, from what is known by ___, to predict ___.

Are we ready for stage 2?

You are ready for the next stage when your frame fits the sentence above and Rohan answered your restate-back with a yes.

Stage 2, requirements: what must the system do, and never do?

Who needs the answer. Rohan and Kalpa's compliance team, who refuse a list that nobody can explain, or that uses data the team may not use, however good its numbers.

The questions on the way. What must the system do? How well? Which limits can we not move? Who may touch it, which data may it use, and who answers for it?

Why does a real team write requirements down?

In November 2019 a customer said his Apple Card credit limit was 20 times his wife's, and a customer service agent could not explain the decision. New York's regulator checked the underwriting data for nearly 400,000 applicants and found no fair lending violation, and the bank explained each decision with ordinary factors once asked (NYDFS report, March 2021). The model was lawful; the requirement nobody wrote down was the explanation and the appeal path.

What does it look like for the gym, and for Kalpa?

The gym's first row is filled. Write all six rows for Kalpa.

Kind	The question to ask	The gym (invented)	Kalpa: our answer, written in pen
Functional (FR): what it must do	What does the user get, and when?	By the 25th, the front desk gets a ranked list of members to call, each with one reason.	
Non-functional (NFR): how well	How fast, how often, how right?		
Constraint: a limit we cannot move	What can we not change?		
Security: who may touch it	Who may see or change the list and the model?		
Privacy: what data may be used	Which data may we not use, or not keep?		
Governance: who answers for it	Who signs it off, and who can stop it?		

Our constraint card on this page
Which row does our card's rule belong in, and what does it add there?

Are we ready for stage 3?

You are ready for the next stage when each of the six rows has a line of your own and your card's rule appears in at least one of them.

Stage 3, options: what are two ways to decide, and which do we choose?

Who needs the answer. Rohan, who pays for a model only if it beats what his team could do with a simple rule, and your evaluator, who presents this sheet on slide 4.

The questions on the way. What is the best simple rule a person could run without a model? What is the model? What does each cost in errors, hours and explanation? Which do we choose, and what fact would flip the choice?

Why does a real team compare two ways?

Under a no-action letter from the US consumer finance regulator, the lender Upstart compared its model with a traditional model on the same applicants for 22 months, with the comparison agreed before any results came in. Upstart reported 27 percent more applicants approved and 16 percent lower average interest rates; the regulator did not replicate the numbers (CFPB, August 2019).

What does it look like for the gym, and for Kalpa?

Fill the error bill row once page 5 has its numbers.

Row of the sheet	Gym A: call members with under 2 visits last month	Gym B: a logistic regression on visits, tenure, plan	Our A: our best simple rule	Our B: our model
What it is	A rule the desk runs on paper	A chance of lapsing for each member		
Error bill per 1,000	470 units	367 units		
Build time	1 hour	1 day		
Can the owner, or Rohan, explain a call?	Yes: "you came once last month"	Yes: the factor that raised the chance most		
How it fails, and how we would notice	Misses members who visit but have decided to leave	Fails quietly if the visit log breaks; the list size jumps		
Our call, and the fact that would flip it	We choose B: it saves 103 units per 1,000 a month	With only 50 calls, the two top-50 lists match and A wins on build time		

Are we ready for stage 4?

You are ready for the next stage when all four of your cells in each row are filled, numbers included, and the last row names one fact, such as a call capacity or a cost ratio, that would change your call.

Stage 4, the POC: does our list beat doing nothing?

Who needs the answer. Rohan's finance team, and your evaluator, who presents slides 4 and 5.

The questions on the way. What do calling nobody and calling everyone cost? Where does the threshold sit? What does our list cost, does it hold on another split, and can we trust the rows?

Why does a real team count an error bill instead of accuracy?

Accuracy prices both mistakes the same, and a lender never does: the German credit data on UCI ships with a cost matrix where passing a bad customer costs 5 and refusing a good one costs 1.

What does it look like for the gym, and for Kalpa?

The gym's test book holds 300 members, 45 of whom lapsed; the desk calls at a chance of 0.2 or more.

	Lapsed	Stayed
Called	30 caught, cost 0	50 wasted, 1 unit each
Not called	15 missed, 4 units each	205 fine, cost 0

The list's bill is 50 x 1 + 15 x 4 = 110 units on 300 members, or 367 per 1,000. The threshold 0.2 is 1 / (1 + 4): call when the chance x 4 is more than (1 minus the chance) x 1.

Number or check	The gym (invented)	Kalpa: our answer, written in pen
Base rate	15 in 100	
Bill of calling nobody, per 1,000	600 units	
Bill of calling everyone, per 1,000	850 units	
Threshold, or list size	0.2	
Our list's bill, per 1,000	367 units	
Our list's bill on five splits, lowest to highest	350 to 390 units	
Does every row describe a different person?	Yes	
Is every value in the data dictionary?	Yes	
Is any group's rate far from the rest?	The student plan lapses twice as often	
Can we trust each column?	No: booked classes counted as visits, so we used check-ins	
Inputs known in time? Test book opened once?	Yes, and yes	

Are we ready for stage 5?

You are ready for the next stage when your list's bill beats calling nobody and calling everyone on the test book, and every member can recompute it by hand from the four counts.

Stage 5, the architecture: what boxes, and one box inside?

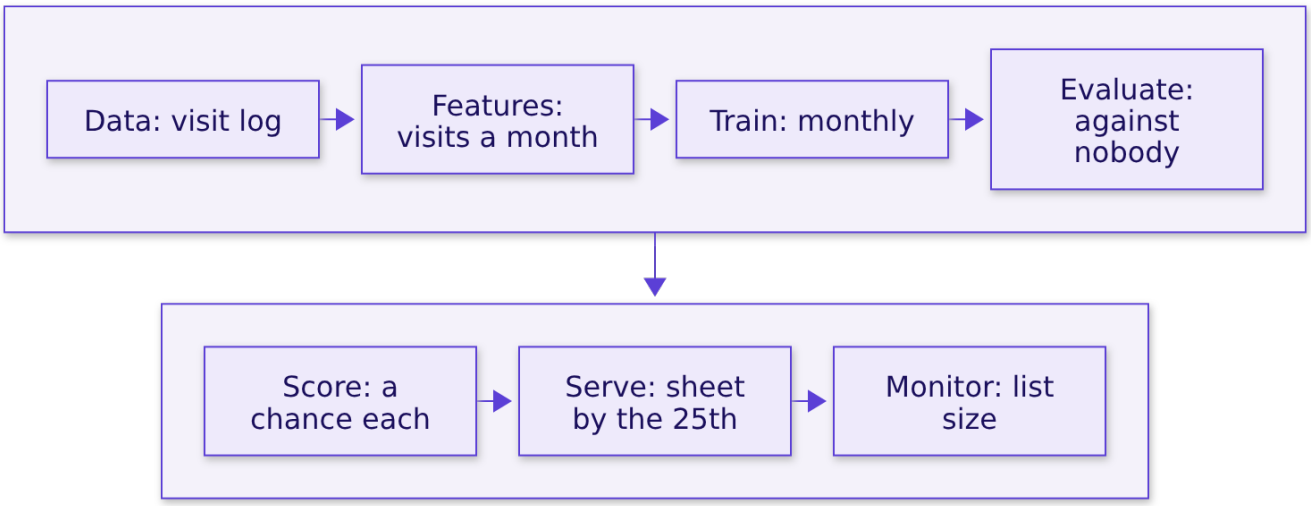
Who needs the answer. Kalpa's engineering lead, who runs the list monthly, and Thursday's gate.

The questions on the way. Which boxes sit inside the black box (the high-level design)? What does one box take in, do and give back (the low-level design)? What if a column arrives late?

Why does a real team draw boxes before it ships?

Stripe's fraud system decides on a payment in under 100 milliseconds (Stripe engineering blog, March 2023); a list built once a month needs none of that speed, so it runs as one batch job.

What does it look like for the gym, and for Kalpa?



Our white box: draw five to seven boxes with arrows, and circle the one we zoom into

Low-level question	The gym's score box (invented)	Our zoomed box, written in pen
What goes in?	The member table on the 24th, one row per member	
What happens inside?	Check columns, load the model, score, keep 0.2 or more, sort, add the reason	
What comes out, and how fast?	A file of member, chance, reason and order, in under a minute	
If a column is late?	The run stops and alerts the owner; no list beats a wrong list	

Are we ready for stage 6?

You are ready for the next stage when a member who did not draw it can point to where the score is computed, who receives the list, and what happens if one column arrives a day late.

Stage 6, the MVP slice: what could Rohan use tomorrow?

Who needs the answer. Rohan, who has to decide what to fund next, and your decision owner, who presents slide 6.

The questions on the way. What did the POC prove, and what can it never prove? What is the smallest thing Kalpa's team could use tomorrow? What does a production system add on top?

Why does a real team separate the POC from the MVP?

Ketan Doshi calls the model "only the tip of the iceberg" and, in his own estimate, puts the work of getting it into production at up to 90 percent of the time and effort (his Enterprise ML blog, 2021). A POC presented as a product hides that work from the client.

What does it look like for the gym, and for Kalpa?

	What the POC proved	What an MVP adds	What production adds
The gym (invented)	On 300 held-out members, the list cost 367 units per 1,000 against 600 for calling nobody.	A script writes the desk's sheet every month, tried for one month at one branch.	It runs itself, alerts when the list size jumps, retrains on new lapses, and a named person can switch it off.
Kalpa: ours			
Kalpa: ours			
Kalpa: ours			

Two lines your plan carries, whatever your card:

1. The Taiwan score proves the pipeline and is never Kalpa's score.
2. This data allows no out-of-time split, so the MVP must add one.

What does our slice hold on Friday?

Piece	Done?
A scoring function: one cardholder in; chance, call or not, main reason and cost out	
A live demo: score one cardholder, show the reason and the cost, change our card's number	
The README, answer first, and the model card	

Are we ready for stage 7?

You are ready for the next stage when a stranger can clone your repository, run one command and see one cardholder's decision, its reason and its cost.

Stage 7, present and defend: which sentence would Rohan act on?

Who needs the answer. Rohan's panel, who will question every member by name, and each of you, since any slide number can be drawn for you to recompute. No AI in the room.

The questions on the way. What is our one sentence, with its numbers? Which questions do we fear most, and who answers each? What would change our call? What is our biggest risk?

Why does a real team defend its numbers in person?

US bank supervisors' model risk guidance of April 2026 (SR 26-2) keeps the idea of effective challenge: critical analysis of a model by objective, qualified and independent people. Saturday's panel is your effective challenge.

What does it look like for the gym?

> "Call the 80 members whose chance of lapsing is 0.2 or more. On last year's members that list cost 367 units per 1,000 against 600 for calling nobody. It was tested on one gym's past members and must be re-tested at the new branch." (invented)

What is our defence for Kalpa?

Stage 7 for Kalpa
Our one sentence, with its numbers
The three questions we fear most, and who answers each
What would change our call
Our top risk from Wednesday's pre-mortem

Are we ready for Saturday?

You are ready when any member, drawn at random, says slide 1 in ninety seconds and recomputes any slide number from the repository in under two minutes.

Which five numbers must I defend alone?

The numbers passport. One copy per member. Fill it alone, in your own words, from your group's own run. No AI.

Name: _____

#	The number	Our value	What it means, in my words	Where it is computed (file and cell)
1	The base rate			
2	The do-nothing bill (calling nobody)			
3	Our threshold or list size			
4	Our list's bill			
5	Our card's number			

The two honesty lines, which I can explain in my own words:

1. The Taiwan score proves the pipeline and is never Kalpa's score.
2. This data allows no out-of-time split, so the MVP must add one.

My recompute drill	Number drawn	Minutes taken	Matched the slide?
First try			
Second try			

Who says what, minute by minute, on Saturday?

The run sheet. Twelve minutes, seven slides, every member speaking. A member of another group holds the timing cards.

Minutes	Slide	Who	What	Our speaker
0:00 to 1:30	1	Drawn by the panel	The recommendation and its error bill in one sentence, then situation, complication, question	Drawn
1:30 to 3:00	2	Decision owner	What each error costs, as one picture, and the metric that follows	
3:00 to 6:00	3	Builder	The live demo: score one customer, show the reason and the cost, change the card's number	
6:00 to 8:00	4	Evaluator	A against B on the error bill, and why we chose	
8:00 to 9:30	5	Evaluator	How honest the number is: the do-nothing bill, the spread, the trap we caught	
9:30 to 11:00	6	Decision owner	What the POC proved and what an MVP still needs (FR, NFR, constraints)	
11:00 to 12:00	7	Builder	What would change our call, the top pre-mortem risk, the ask	

Each day closes with the opener: one member of every group, drawn, says slide 1 to the room in ninety seconds, timed by another group. There is no score.

Rehearsal	Total time	Slide that ran over, and by how much	What we cut
Dry run 1, Thursday, with our red-team partner			
Dry run 2, Friday, cold and recorded			

What surprised us, and what did we decide?

The challenges log. One line per decision or surprise, in the group's own words. Log every sealed hint you open.

Day	Stage	What surprised us or went wrong	What we decided, and why	Hint opened	Logged by

Where did AI write code for us, and who can explain it?

The AI log. One line for every use of AI. The explainer is a member who did not write the prompt, and explains the code line by line before it is committed.

Day	Stage	Tool	What we asked	What we kept	How we checked it	Explained by